

Chapter 4 Experimental Results Revision Pack

This revision pack is based on the generated experiment outputs in Fingerprint_Enhancement_Project/outputs/final_mode_a/script/. The editable report source was not available in the workspace; only the compiled PDF was found. Therefore, the following text is prepared as direct replacement and insertion content for Chapter 4.

1. Revised Configuration Paragraph

Replace the existing paragraph beginning with "Table 4.1 summarises the selected configuration..." and ending with "...were not modified using Validation Set results." with:

Table 4.1 summarises the selected configuration for each enhancement technique after Development Set parameter searching. NLM selected a denoising strength of $h = 0.07$, TV Restoration selected a regularisation weight of 0.02, and Coherence-Guided Directional Diffusion selected 2 iterations with a step size of 0.12. The final Modified Gabor configuration used $\text{detail_clip} = 0.10$, frequencies = [0.075, 0.095, 0.115, 0.140], $\gamma = 0.50$, $\text{orientation_bins} = 12$, $\text{orientation_offset} = 1.5707963267948966$ radians, $\sigma_{\text{scale}} = 0.55$, and $\text{strength} = 0.12$. All selected configurations were finalised before validation and were not modified using Validation Set results.

2. Corrected Table 4.1

Table 4.1: Development-Selected Configuration for Each Enhancement Technique

Member	Technique	Development-Selected Configuration	Selection Status
M1	NLM	$h = 0.07$	Selected
M2	Modified Gabor	$\text{detail_clip} = 0.10$; frequencies = [0.075, 0.095, 0.115, 0.140]; $\gamma = 0.50$; $\text{orientation_bins} = 12$; $\text{orientation_offset} = 1.5707963267948966$ radians; $\sigma_{\text{scale}} = 0.55$; $\text{strength} = 0.12$	Selected
M3	TV Restoration	$\text{weight} = 0.02$	Selected
M4	Coherence-Guided Directional Diffusion	iterations = 2; $\text{step} = 0.12$	Selected

3. Corrected Development Results Table

Replace the existing Table 4.2 with:

Table 4.2: Quantitative Performance Comparison on the Development Dataset

Method	Images	Mean PSNR	Mean SSIM	Mean MSE	Delta PSNR	Delta SSIM	Mean Runtime (s/image)
Degraded Baseline	500	27.357	0.8415	0.001842	–	–	–
M1 NLM	500	29.568	0.9336	0.001114	2.2110	0.0921	0.259
M2 Modified Gabor	500	27.367	0.8419	0.001838	0.0102	0.0004	0.468
M3 TV	500	29.287	0.9379	0.001188	1.9303	0.0964	0.239
M4 Directional Diffusion	500	27.986	0.8511	0.001593	0.6296	0.0096	0.528

4. Revised Development Results Discussion

Replace the existing development-results discussion beginning with "Table 4.2 presents the quantitative comparison..." and ending with "...used in the experiment." with:

Table 4.2 presents the quantitative comparison on the 500-image Development Set. NLM achieved the highest mean PSNR of 29.568 dB, improving PSNR by approximately 2.2110 dB over the degraded baseline. It also achieved a strong mean SSIM of 0.9336 and the lowest mean MSE of approximately 0.001114 among the four enhancement techniques. TV Restoration achieved the highest mean SSIM of 0.9379 and recorded the fastest mean runtime among the enhancement methods at approximately 0.239 seconds per image. Modified Gabor produced only a very small quantitative improvement over the degraded baseline, while Coherence-Guided Directional Diffusion produced a moderate improvement but remained below NLM and TV in overall quantitative performance.

5. New Validation Results Table

Insert the following table immediately after the revised development-results discussion. Since no later tables were present in Chapter 4, this should be numbered Table 4.3.

Table 4.3: Quantitative Performance Comparison on the Validation Dataset

Method	Images	Mean PSNR	Mean SSIM	Mean MSE	Delta PSNR	Delta SSIM	Mean Runtime (s/image)
Degraded Baseline	1000	27.363	0.8385	0.001838	–	–	–
M1 NLM	1000	29.559	0.9300	0.001114	2.1962	0.0915	0.272
M2 Modified Gabor	1000	27.373	0.8388	0.001834	0.0096	0.0004	0.451
M3 TV	1000	29.321	0.9358	0.001177	1.9582	0.0973	0.230
M4 Directional Diffusion	1000	27.992	0.8481	0.001590	0.6293	0.0096	0.508

6. New Validation Results Discussion

Insert immediately after Table 4.3:

The validation results are highly consistent with the Development Set findings. NLM achieved the highest validation PSNR of 29.559 dB, corresponding to an improvement of 2.1962 dB compared with the degraded validation baseline, and maintained a low mean MSE of 0.001114. TV Restoration achieved the highest validation SSIM of 0.9358, produced the largest SSIM improvement of approximately 0.0973, and was also the fastest enhancement method on the Validation Set at approximately 0.230 seconds per image. Directional Diffusion required approximately 0.508 seconds per image, while Modified Gabor again produced only minimal quantitative improvement. The similarity between the Development and Validation Set results indicates that the relative performance of the methods remained stable on unseen fingerprint images.

7. Revised Overall Comparison / Winner Paragraph

Replace the first two paragraphs of Section 4.2, from "Based on the predefined Development Set ranking rule..." through "...in terms of SSIM and runtime." with:

Based on the predefined PSNR-first ranking criterion, Non-Local Means (NLM) was selected as the overall best-performing enhancement method. It achieved the highest PSNR and lowest MSE on both the Development and Validation Sets while maintaining high SSIM values. On the Development Set, NLM achieved a mean PSNR of 29.568 dB and the lowest mean MSE of 0.001114. Its performance remained stable on the independent Validation Set, where it achieved a mean PSNR of 29.559 dB and the same rounded mean MSE of 0.001114. This indicates that the selected NLM configuration generalised well to fingerprint images from unseen subjects rather than only performing well on the Development Set.

Nevertheless, Total Variation (TV) Restoration achieved the highest SSIM and the fastest runtime on both datasets. On the Validation Set, TV achieved a mean SSIM of 0.9358 compared with 0.9300 for NLM, and required approximately 0.230 seconds per image. Therefore, TV is particularly strong in structural preservation and computational efficiency, while NLM provides the strongest overall reconstruction quality according to the selected PSNR-first evaluation criterion. The selection of NLM therefore reflects the weighting of the adopted evaluation criteria rather than superiority across every individual metric.

8. Corrected Directional Diffusion Runtime Paragraph

Replace the paragraph beginning with "Coherence-Guided Directional Diffusion produced a moderate improvement..." and ending with "...used in the current experiment." with:

Coherence-Guided Directional Diffusion produced a moderate improvement over the degraded baseline but required the highest processing time among the four methods. On the Validation Set, it improved mean PSNR from 27.363 dB to 27.992 dB but required approximately 0.508 seconds per image. This suggests that its structure-aware iterative processing introduces additional computational cost while providing less benefit than NLM and TV under the Gaussian-noise degradation used in the current experiment.

9. New Structural Evaluation Subsection

Insert the following subsection after the paragraph that explains why altered fingerprints are kept separate from the main paired PSNR, SSIM, and MSE experiment. Number it as Table 4.4.

Structural Evaluation on Altered Fingerprint Images

Since the altered fingerprint subsets are not treated as pixel-aligned clean reference pairs, their evaluation focuses on structural ridge measurements rather than full-reference PSNR, SSIM, and MSE. Table 4.4 summarises the overall structural results across 1,500 altered fingerprint images from the Altered-Easy, Altered-Medium, and Altered-Hard subsets.

Table 4.4: Structural Evaluation on Altered Fingerprint Images

Method	Altered Images	Mean Coherence	Mean Contrast	Mean Fragmentation	Mean Continuity	Mean Ridge Coverage
Altered Baseline	1500	0.6508	0.1826	5.1562	22.3965	0.3693
M1 NLM	1500	0.7234	0.1740	3.7403	28.2292	0.3617
M2 Modified Gabor	1500	0.6516	0.1828	5.1535	22.4020	0.3693
M3 TV	1500	0.6676	0.1742	4.6801	23.1128	0.3657
M4 Directional Diffusion	1500	0.6541	0.1777	5.1788	22.3086	0.3668

NLM demonstrated the strongest overall structural improvement on the altered fingerprint evaluation. It increased mean coherence from 0.6508 to 0.7234 and mean continuity from 22.3965 to 28.2292, while also reducing mean fragmentation from 5.1562 to 3.7403. TV Restoration also improved coherence, continuity, and fragmentation, but the magnitude of its structural improvement was smaller than that of NLM. Modified Gabor produced only minimal structural change, while Directional Diffusion showed a small coherence increase but did not improve continuity or fragmentation overall. This structural evaluation is useful because traditional full-reference metrics such as PSNR, SSIM, and MSE measure similarity to a reference image, whereas altered-image structural evaluation provides additional evidence about how well fingerprint ridge patterns are preserved or restored under more difficult image degradation.

10. Replacement / Insertion Checklist

- Replace the Table 4.1 introductory configuration paragraph with the revised configuration paragraph in Section 1.
- Replace the existing Table 4.1 row for Modified Gabor, or replace the full Table 4.1 with the corrected version in Section 2.
- Replace the existing Table 4.2 with the corrected Development Dataset table

in Section 3.

- Replace the outdated development-results discussion that reports NLM mean MSE as approximately 0.001121 with the revised paragraph in Section 4.
- Insert the new Validation Dataset table as Table 4.3 after the development results discussion.
- Insert the validation-results discussion in Section 6 immediately after Table 4.3.
- Replace the Section 4.2 overall winner discussion with the revised two paragraphs in Section 7.
- Replace the outdated TV validation runtime sentence reporting approximately 0.254 seconds per image with the corrected value of approximately 0.230 seconds per image, as included in Section 7.
- Replace the outdated Directional Diffusion validation runtime sentence reporting approximately 0.772 seconds per image with the corrected paragraph in Section 8.
- Insert the structural evaluation subsection and Table 4.4 in Section 9 after the paragraph explaining that altered fingerprints are not used for paired full-reference metrics.
- No figure renumbering is required because only tables are added. The newly added validation and structural tables should be numbered Table 4.3 and Table 4.4 respectively.